

# Aaditya R. Bhat - EE/CS10b Extra Credit #2

## Code Optimization (30 points)

For this assignment there is **no collaboration or use of AI/LLMs allowed**

For the following code fragments, rewrite or write them to minimize the number of lines and/or bytes of code. That is, the number of lines of code and/or the number of bytes the assembled code generates. The operation of the code is correct (there are no errors) and must not be changed, nor may any additional registers be used or changed (except additional registers may be used in problem #4). The code is only required to have the effects listed in the comments. Other "side effects" do not need to be replicated and no new "side effects" should occur. Be sure to test the code syntax in the assembler.

Assume each problem is independent and assume `Buffer` is the label of a byte in the data segment and `WIN_DIGIT` is a constant between 0 and 7.

### 1) Negation (5 points)

```
;OLD

;R16 <- -R16
COM R16
INC R16

;NEW

;R16 <- -R16
NEG R16
```

### 2) Reading Memory (5 points)

```
;OLD

;R16 <- value stored at [Buffer + WIN_DIGIT]
LDI YL, LOW(Buffer)
LDI YH, HIGH(Buffer)
LDI R17, WIN_DIGIT
LDI R18, 0
ADD YL, R17
ADC YH, R18
LD R16, Y

;NEW

;Since Buffer is fixed label and WIN_DIGIT is a constant, assembler
;can calculate Buffer + WIN_DIGIT

LDS R16, Buffer + WIN_DIGIT
```

### 3) Set SREG bit (5 points)

```
;OLD

;set bit 7 in SREG
IN R16, SREG
ORI R16, 0b10000000
OUT R16, SREG

;NEW

;7th Bit is interrupt flag
SEI
```

### 4) Shifting R17 | R16 by 4 bits (15 points)

Given: R17 | R16 - 16-bit unsigned value (FEDCBA98 | 76543210) to shift right by 4 bits

Result: R17 | R16 shifted right by 4 bits (xxxxFEDC | BA987654, the upper nibble of R17 can be any value)

Code: must use fewer than 8 instructions

```
;Given

LDI R17, FEDCBA98
LDI R16, 76543210

;Implementation

(1) SWAP R17          ;R17 = xxxxFEDC Done
(2) MOV  R18, R17      ;R18 = BA98FEDC
(3) ANDI R18, $F0      ;R18 = BA980000
(4) SWAP R16          ;R16 = 32107654
(5) ANDI R16, $0F      ;R16 = 00007654
(6) OR   R16, R18      ;R16 = BA987654 Done
```